

Chapter Two

Bohr Quantization Rule for Hydrogen Atom

1.2 Bohr Quantization Rule for Hydrogen Atom

Bohr give classical model of hydrogen atom in which electron moving in a circular orbit of radius r has an angular momentum mvr which is constant. The generalized coordinate for is assume to be θ which vary from $0 \leq \theta \leq 2\pi$ and conjugate momentum is p_θ which is angular momentum L_z (we assume motion is confine into $x-y$ plane so angular momentum is in z direction)

According to Bohr quantization rule

$$\oint p_\theta dq = n_q h \Rightarrow \oint p_\theta d\theta = nh \Rightarrow \oint L_z d\theta = nh \text{ where } n = 1, 2, 3 \dots$$

$$L_z \int_0^{2\pi} d\theta = nh \Rightarrow mvr = \frac{nh}{2\pi} \Rightarrow mvr = n\hbar \text{ which is very famous Bohr quantization rule .}$$

De Broglie wavelength and hydrogen atom

From Bohr quantization rule $mvr = n\hbar$

$$pr = n\hbar \text{ from de-Broglie hypothesis } p = \frac{h}{\lambda}$$

$$\frac{h}{\lambda} r = n\hbar = \frac{nh}{2\pi} \Rightarrow 2\pi r = n\lambda \text{ Where } n = 1, 2, 3 \dots$$

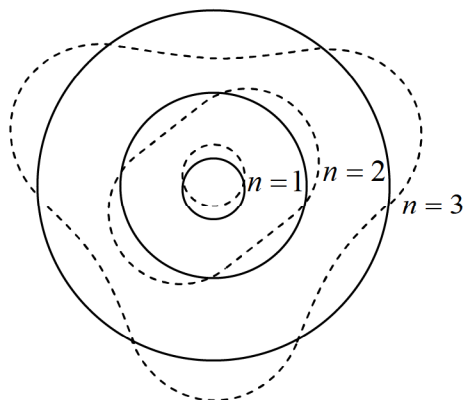


Figure 7.5

The allowed orbits are those in which the circumference of the orbit can contain exactly an integral number of de Broglie wavelengths. It can be imagined that electron is moving in a circular orbit at constant speed with the associated wave following the electron. The wave of wavelength λ is wrapped repeatedly around the circular orbit.

In the above figure de-Broglie standing wave interpretation satisfies Bohr's quantization rule.

Bohr Model of Hydrogen Atom

In atomic physics, the **Bohr model**, introduced by Niels Bohr in 1913, depicts the atom as small, with a positively charged nucleus surrounded by electrons that travel in circular orbits around the nucleus—similar in structure to the solar system, but with attraction provided by electrostatic forces rather than gravity.

He suggested that electrons could only have certain **classical** motions:

1. Electrons in atoms orbit the nucleus which consists of single positive charge particle proton.

There is only external force action on electron is electrostatic force.

2. The electrons can only orbit stably, without radiating, in certain orbits (called by Bohr the "stationary orbits"): at a certain discrete set of distances from the nucleus. These orbits are associated with definite energies and are also called energy shells or energy levels. In these orbits, the electron's acceleration does not result in radiation and energy loss as required by classical electromagnetism.

The significance of the Bohr model is that the laws of classical mechanics apply to the motion of the electron about the nucleus *only when restricted by a quantum rule*. The angular momentum $L_z = mvr = n\hbar$ is restricted to be an integer multiple of a fixed unit:

$$L_z = n\hbar \text{ where } n = 1, 2, 3 \dots \text{ and}$$

n is called the principle quantum number, and $h = \frac{h}{2\pi}$. The lowest value of n is 1. This gives a

smallest possible orbital radius of $0.592 \text{ \AA}$ known as the **Bohr Radius**.

3. Electrons can only gain and lose energy by jumping from one allowed orbit to another, absorbing or emitting electromagnetic radiation with a frequency ν determined by the energy difference of the levels according to the *Planck relation*:

$$\Delta E = E_f - E_i = h\nu$$

where h is Planck's constant. The frequency of the radiation emitted at an orbit of period T is as it would be in classical mechanics; it is the reciprocal of the classical orbit period: $\nu = \frac{1}{T}$

Hydrogen Atom

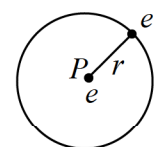
A **hydrogen atom** is an atom of the chemical element hydrogen. The electrically neutral atom contains a single positively charged proton and a single negatively charged electron bound to the nucleus by the Coulomb force. According to Bohr electron revolve about the nucleus in different quantized circular orbits whose angular momentum is given by $L = n\hbar$ where $n = 1, 2, 3, \dots$

The electron is held in a circular orbit by electrostatic attraction. The centripetal force is equal

to the Coulomb force. $\frac{m_e v^2}{r} = \frac{k_e e^2}{r^2}$

where m_e and e the electron's mass and charge of the electron,

$k_e = \frac{1}{4\pi\epsilon_0}$ is identified as Coulomb's constant.



If v is velocity of electrons in orbit, the electron's speed at any radius:

$$v = \sqrt{\frac{k_e e^2}{m_e r}} \Rightarrow v = \sqrt{\frac{e^2}{m_e 4\pi\epsilon_0 r}}$$

The electron's total energy at any radius: $E = T + V = \frac{m_e v^2}{2} - \frac{k_e e^2}{r}$

Putting the value of v one will get $E = -\frac{k_e e^2}{2r}$. The total energy is half the potential energy.

From the quantization the angular momentum $L_z = n\hbar = m_e v r = n\hbar$

Substituting the expression for the velocity gives an equation for r in terms of n :

$$\sqrt{k_e e^2 m_e r} = n\hbar \text{ allowed orbit radius at any } n \text{ is: } r_n = \frac{n^2 \hbar^2}{k_e e^2 m_e} \Rightarrow r_n = \frac{4\pi\epsilon_0 n^2 \hbar^2}{m_e e^2}$$

For $n=1$ $r_1 = a_0 = 0.53 \times 10^{-10} m$ called the Bohr radius.

Hence radius of hydrogen atom is quantized one can also see speed of electron is also quantized

$$v = \sqrt{\frac{e^2}{m_e 4\pi\epsilon_0 r}} \Rightarrow \sqrt{\frac{e^4}{(4\pi\epsilon_0)^2 n^2 \hbar^2}} = \frac{e^2}{4\pi\epsilon_0 n \hbar} = \frac{2.2 \times 10^6 m/sec}{n}$$

The energy of the n^{th} level for any atom is determined by the radius and quantum number:

$$E_n = -\frac{k_e e^2}{2r_n} = -\frac{(k_e e^2)^2 m_e}{2\hbar^2 n^2} = -\frac{m_e}{2\hbar^2} \left(\frac{e^2}{4\pi\epsilon_0} \right)^2 \frac{1}{n^2} = \frac{-13.6}{n^2} eV$$

The Rydberg energy R_E which is given by $R_E = \frac{(k_e e^2)^2 m_e}{2\hbar^2} \Rightarrow \frac{m_e}{2\hbar^2} \left(\frac{e^2}{4\pi\epsilon_0} \right)^2$

1.3 Hydrogen Like Atom

A **hydrogen-like atom/ion** (usually called a "hydrogenic atom") is any atomic nucleus bound to one electron and thus is is electronic with hydrogen. These atoms or ions can carry the positive charge where is the atomic number of the atom. Examples of hydrogen-like atoms/ions are Hydrogen itself, He^+ , Li^{2+} , Be^{3+} and B^{4+} . Because hydrogen-like atoms/ions are two-particle systems with an interaction depending only on the distance between the two particles.

A correction in radius velocity and energy due to atomic number Z of atom

If the atomic number of hydrogen like atom is Z then potential energy of system will

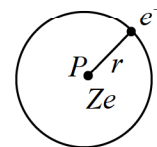
$$V(r) = -\frac{q_1 q_2}{4\pi\epsilon_0 r^2} = -\frac{Ze.e}{4\pi\epsilon_0 r^2}. \text{ Accordingly there is correction in physical measurement of}$$

hydrogen like Atom.

$$1 \text{ Correction in radius } r_n = \frac{4\pi\epsilon_0 n^2 \hbar^2}{m_e Ze.e} = \frac{4\pi\epsilon_0 n^2 \hbar^2}{m_e Ze^2} = \frac{0.53 n^2}{Z} A^0$$

$$2 \text{ Correction in speed } v_n = \frac{Ze.e}{4\pi\epsilon_0 n \hbar} = \frac{Z}{n} \times 2.2 \times 10^6 m/sec$$

$$3 \text{ Correction in energy } E_n = -\frac{m_e}{2\hbar^2} \left(\frac{Ze.e}{4\pi\epsilon_0} \right)^2 \frac{1}{n^2} = -Z^2 \frac{13.6}{n^2}$$



Correction Due to Finite Nuclear Mass

In Physical measurement of hydrogen atom it is assumed that mass of nucleus is infinite in comparison of electron. But if mass of nucleus is finite then the system is two particle system in

which the orbiting mass is replaced by reduce mass $\mu = \frac{m_n \cdot m_e}{m_n + m_e}$ where m_n is mass of nucleus

and m_e mass of electron. (The derivation is in equivalent to two body planetary motion discussed in mechanics)

$$1. \text{ Correction in radius } r_n = \frac{4\pi\epsilon_0 n^2 \hbar^2}{\mu Z e^2} = \frac{m_e 4\pi\epsilon_0 n^2 \hbar^2}{\mu m_e Z e^2} = 0.53 \frac{m_e}{\mu} \frac{n^2}{Z} A^0$$

$$2. \text{ Correction in speed } v_n = \frac{Z e e}{4\pi\epsilon_0 n \hbar} = \frac{Z}{n} \times 2.2 \times 10^6 \text{ m/sec no Correction due to reduce mass}$$

$$3. \text{ Correction in energy } E_n = -\frac{\mu}{2\hbar^2} \left(\frac{Z e e}{4\pi\epsilon_0} \right)^2 \frac{1}{n^2} = -\frac{\mu}{m_e} \frac{m_e}{2\hbar^2} \left(\frac{Z e e}{4\pi\epsilon_0} \right)^2 \frac{1}{n^2} = -Z^2 \frac{\mu}{m_e} \frac{13.6}{n^2}$$

Relationship between kinetic energy T potential energy U and total energy E

$$U = -\frac{k_e e^2}{r}, T = \frac{k_e e^2}{2r}, E = -\frac{k_e e^2}{2r}$$

$$U = \frac{E}{2}, T = -E$$

The spectrum of hydrogen atom

The electron may undergo a transition to some other stationary state either by absorbing energy and moving up to a higher energy state or giving off energy and moving down. The transition between energy levels are constantly occurring. The atom absorbed or released energy in form of electromagnetic radiation.

Energy of radiation corresponds to difference of energy between transition levels initially in n_i to final level n_f

$$\text{So } E_\gamma = E_f - E_i = \left(\frac{-13.6}{n_f^2} \right) - \left(\frac{-13.6}{n_i^2} \right) = 13.6 \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right)$$

If E_γ is positive means $n_f > n_i$ mean hydrogen atom absorb radiation

If E_γ is positive means $n_f < n_i$ mean hydrogen atom release radiation

The wavelength of radiation and Rydberg constant

According to Planck formula the energy of radiation is given by $E_\gamma = h\nu = \frac{hc}{\lambda}$

$$E_\gamma = |E_f - E_i| = \frac{m_e}{2\hbar^2} \left(\frac{e^2}{4\pi\epsilon_0} \right)^2 \left| \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right) \right|$$

$$\frac{hc}{\lambda} = \frac{2\pi\hbar c}{\lambda} = \frac{m_e}{2\hbar^2} \left(\frac{e^2}{4\pi\epsilon_0} \right)^2 \left| \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right) \right| \Rightarrow \frac{1}{\lambda} = \frac{m_e}{4\pi c \hbar^3} \left(\frac{e^2}{4\pi\epsilon_0} \right)^2 \left| \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right) \right|$$

$$\frac{1}{\lambda} = R_\infty \left| \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right) \right| \text{ where } R_\infty = \frac{m_e}{4\pi c \hbar^3} \left(\frac{e^2}{4\pi\epsilon_0} \right)^2 = 1.097 \times 10^7 \text{ m}^{-1}$$

known as Rydberg constant R_∞

Correction in Rydberg constant for hydrogen like atom

If Z is atomic number and μ is the reduce mass of hydrogen like atom the

$$R_\mu = \frac{\mu Z^2}{4\pi c \hbar^3} \left(\frac{e^2}{4\pi\epsilon_0} \right)^2 = \frac{\mu}{m_e} \cdot Z^2 \times R_\infty \Rightarrow R_\mu = \frac{\mu}{m_e} \cdot Z^2 \times 1.097 \times 10^7 \text{ m}^{-1}$$

Spectral Series

When an electric discharge is passed through gaseous hydrogen, the H_2 molecules dissociate and the energetically excited H atoms that are produced emit light of discrete frequencies, producing a spectrum of a series of 'lines'.

(i) **Lyman Series:** $\frac{1}{\lambda} = R \left(\frac{1}{1^2} - \frac{1}{n^2} \right)$;

$n = 2, 3, 4 \dots$ (In U.V. region), where, R is Rydberg constant ($1.097 \times 10^7 \text{ m}^{-1}$)

(ii) **Balmer Series:**

$$\frac{1}{\lambda} = R \left(\frac{1}{2^2} - \frac{1}{n^2} \right) \quad n = 3, 4, 5 \dots \text{ (in visible region), } n = 3 \text{ for } H_\alpha \text{ line } n = 4 \text{ } H_\beta \text{ line}$$

$n = 5 \text{ } H_\gamma \text{ line}$

(iii) **Paschen Series:**

$$\frac{1}{\lambda} = R \left(\frac{1}{3^2} - \frac{1}{n^2} \right) \quad n = 4, 5, 6 \dots \text{ (Near Infra Red)}$$

(iv) **Bracket Series:**

$$\frac{1}{\lambda} = R \left(\frac{1}{4^2} - \frac{1}{n^2} \right); n = 5, 6, 7 \dots \text{ (Infra Red)}$$

(v) **Pfund Series:** $\frac{1}{\lambda} = R \left(\frac{1}{5^2} - \frac{1}{n^2} \right); n = 6, 7, 8 \dots \text{ (Far Infra Red)}$

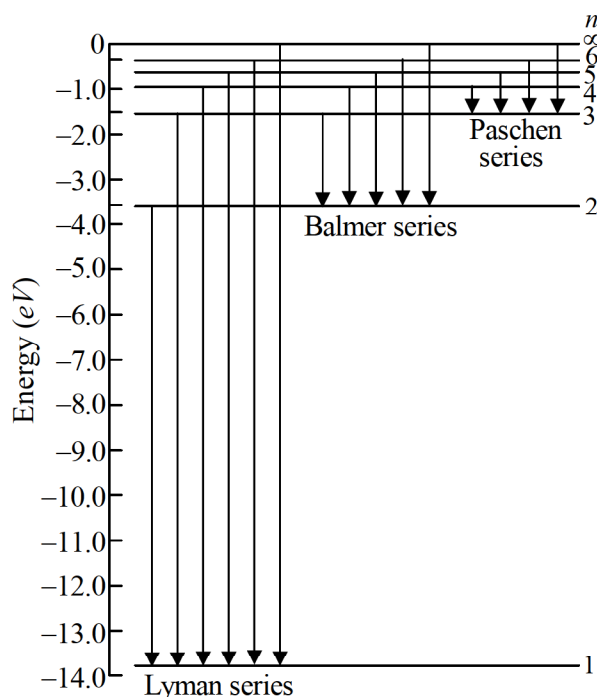


Figure 7.6

Ionization energy of hydrogen atom: If electron is in ground state $n=1$, then Minimum energy required by hydrogen atom such that electron became unbounded. Electron became unbounded it will be in continuum state which has energy $E \geq 0$

Initial state is $n=1$, $E_1 = -\frac{13.6}{1^2} = -13.6\text{eV}$ the final state is $n = \infty$ $E_f = -\frac{13.6}{\infty} = 0$

ΔE is Energy required by atom such that electron will transit from $E_1 \rightarrow E_f$

$$\Delta E = E_f - E_1 = 0 - (-13.6\text{eV}) = 13.6\text{eV}$$

Example: If hydrogen atom at rest it emits a photon in going from $n=5$ state to $n=1$ state is m/s . Find the recoil speed of a hydrogen atom in meter/sec after the transition

Solution: If Momentum of recoil hydrogen atom $p_H = mv$ where mass of hydrogen atom $m = 1.67 \times 10^{-27} \text{ kg}$

Momentum of emitted photon $P_\gamma = \frac{\Delta E}{c}$

Linear momentum is conserved in the recoil process so $P_H = P_\gamma$

$$\Delta E = E_5 - E_1 = -13.6 \left[\frac{1}{5^2} - \frac{1}{1^2} \right] \text{eV} = (13.6) \left(\frac{24}{25} \right) \text{eV} = \frac{13.6 \times 24}{25} \times (1.6 \times 10^{-19}) \text{ J}$$

$$\Rightarrow \Delta E = 2.1 \times 10^{-18} \text{ J}$$

$$P_H = P_\gamma \Rightarrow mv = \frac{\Delta E}{c} \quad \text{or} \quad v = \frac{\Delta E}{mc} \Rightarrow v = \frac{2.1 \times 10^{-18}}{(1.67 \times 10^{-27}) \times (3 \times 10^8)} = 4.19 \text{ m/s}$$

Example: A single electron orbits around a stationary nucleus of charge $+Ze$, where Z is a constant and e is the magnitude of the electronic charge. It requires 47.2eV to excite the electron from the second Bohr orbit to the third Bohr orbit. Find

- (a) The value of Z .
- (b) The energy required to excite the electron from the third to the fourth Bohr orbit
- (c) The wavelength Of the electromagnetic radiation required to remove the electron from the first Bohr orbit to infinity.
- (d) The kinetic energy, potential energy, and the angular momentum of the electron in the first Bohr orbit.
- (e) The radius of the first Bohr orbit.

(The ionization energy of hydrogen atom $= 13.6\text{eV}$, Bohr radius $= 5.3 \times 10^{-11}$ meter, velocity of light $= 3 \times 10^8 \text{ m/sec}$. Planck's constant $= 6.6 \times 10^{-34} \text{ Joule - sec}$)

Solution: (a) According to Bohr model, for hydrogen like atom $E_n = -\frac{13.6Z^2}{n^2} eV$

$$\therefore (E_3 - E_2) = -13.6Z^2 \left[\frac{1}{9} - \frac{1}{4} \right] \text{ or } 47.2 = +\frac{13.6Z^2 \times 5}{36} \Rightarrow Z^2 = \frac{47.2 \times 36}{13.6 \times 5} = 25 \text{ or } Z = 5$$

$$(b) E_4 - E_3 = -13.6Z^2 \left[\frac{1}{4^2} - \frac{1}{3^2} \right] = \frac{13.6Z^2 \times 7}{9 \times 16} = \frac{13.6 \times 25 \times 7}{9 \times 16} = 16.53 eV$$

$$(c) \lambda \text{ for removing electron from } n=1 \text{ to } n=\infty \because E_\infty = 0 \quad E = -E_1 = +\frac{13.6}{(1)^2} \times Z^2 eV$$

$$E = 13.6 \times 25 \times 1.6 \times 10^{-19} J \Rightarrow \frac{hc}{\lambda} = 13.6 \times 25 \times 1.6 \times 10^{-19}$$

$$\Rightarrow \lambda = \frac{(6.6 \times 10^{-34}) \times (3 \times 10^8)}{13.6 \times 25 \times 1.6 \times 10^{-19}} \Rightarrow \lambda = 3.64 \times 10^{-9} m$$

$$(d) \text{ Total energy in first orbit} = -\frac{13.6Z^2}{n^2}; \quad E = -\frac{13.6 \times 25}{1} = -340 eV$$

$$E = -340 eV \quad T = -E \text{ or } T = 340 eV$$

$$U = 2E = 2(-340) eV \Rightarrow U = -680 eV, \quad L_z = \frac{nh}{2\pi} = \frac{1 \times 1.66 \times 10^{-34}}{2\pi} = 1.05 \times 10^{-34} Js$$

$$(e) r_1 = \frac{\text{Bohr radius}}{Z} \quad \text{or } \Rightarrow r_1 = \frac{5.3 \times 10^{-11}}{5} \Rightarrow r_1 = 1.06 \times 10^{-11} m$$

Example: A double ionized lithium atom is hydrogen-like with atomic number 3.

(a) Find the wavelength of the radiation required to excite the electron in Li from the first to the third Bohr orbit. (Ionization energy of the hydrogen atom equal $13.6 eV$).

(b) How many spectral lines are observed in the emission spectrum of the above excited system?

Solution: (i) For hydrogen like atom $E_n = \left(-\frac{13.6}{n^2} Z^2 \right) eV$ λ of radiation excite electron from $n=1$ to $n=3$:

$$\text{Given: } Z = 3, \quad E_1 = -\frac{13.6}{(1)^2} \times (3)^2 = -13.6 \times 9 eV \text{ and } E_3 = -\frac{13.6}{(3)^2} \times (3)^2 = -13.6 eV$$

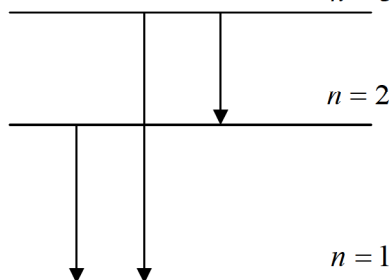
$$\therefore E_3 - E_1 = -13.6 - (-13.6 \times 9) \quad \Delta E = 13.6 \times 8$$

$$\text{or } \Delta E = 108.8 eV = 108.8 \times 1.6 \times 10^{-19} J \Rightarrow \frac{hc}{\lambda} = (108.8 \times 1.6 \times 10^{-19})$$

$$\lambda = \frac{(6.6 \times 10^{-34}) \times (3 \times 10^8)}{108.8 \times 1.6 \times 10^{-19}} \Rightarrow \lambda = 1.137 \times 10^{-8} m = 113.7 \text{ \AA}$$

(b) Number of spectral lines in emission spectrum:

or Number of spectral lines = $\frac{n(n-1)}{2}$ Number of spectral lines = $\frac{3 \times 2}{2}$
 $n = 3$



$\therefore$ Number of spectral lines = 3. The transition are $3 \rightarrow 1, 3 \rightarrow 2, 2 \rightarrow 1$

Example: A muonic atom contains a nucleus of charge Ze and a negative muon, μ^- , moving about it. The μ^- is an elementary particle with charge $-e$ and a mass that is 207 times as large as an electron mass.

- Calculate the muon-nucleus separation, D , of the first Bohr orbit of a muonic atom with $Z = 1$.
- Calculate the binding energy of a muonic atom with $Z = 1$
- What is the wavelength of the first line in the Lyman series for such an atom?

Solution: (a) The reduced mass of the system, with $m_{\mu^-} = 207m_e$ and $M = 1836m_e$, is, from (4-20)

$$\mu = \frac{207m_e \times 1836m_e}{207m_e + 1836m_e} = 186m_e$$

Then, from (4-16), with $n = 1, Z = 1$ and $m = 186m_e$, we obtain

$$D_1 = \frac{4\pi\epsilon_0 \hbar^2}{186m_e e^2} = \frac{1}{186} \times 5.3 \times 10^{-11} m = 2.8 \times 10^{-13} m = 2.8 \times 10^{-3} \text{ \AA}$$

Therefore the μ^- is much closer to the nuclear (proton) surface than is the electron in a hydrogen atom. It is this feature which makes such muonic atoms interesting, information about nuclear properties being revealed from their study.

(b) From (4-18) with $Z = 1, n = 1$ and $m = \mu = 186m_e$, we have

$$E = -186 \frac{m_e e^4}{(4\pi\epsilon_0)^2} = -186 \times 13.6 eV = -2530 eV$$

as the ground state energy. Hence, the binding energy is $2530 eV$.

(c) From (4-22), with $Z = 1$, we have

$$\kappa = R_M \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

For the first Lyman line, $n_i = 2$ and $n_f = 1$. In this case, $R_M = (\mu / m_e) R_\infty = 186 R_\infty$. Hence

$$\kappa = \frac{1}{\lambda} = 186 R_\infty \left(1 - \frac{1}{4} \right) = 139.5 R_\infty$$

With $R_\infty = 109737 \text{ cm}^{-1}$ $\lambda \approx 6.5 \text{ \AA}$

so that the Lyman lines lie in the x - ray part of the spectrum. X - ray techniques are necessary, therefore, to study the spectrum of muonic atoms.